

Homework 1

Math 2031 — Professor Hamilton

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Section 1.1

Question 4

- (A) $\exists$ triangle T such that T is not an isosceles triangle.
- (B) $\forall$ even numbers N , there does not exist an $A \in \mathbb{R}$ such that $N = 3A$.
- (C) There exists a door in the building that was not locked.
- (D) There exists a new car such that there is nothing wrong with it.
- (E) All angles of a triangle are less than or equal to 90 degrees.
- (F) $\forall$ sets S , S does not contain infinitely many elements.

Question 5

- (A) $\forall x \in \mathbb{R}, x^2 + x + 1 \neq 0$.
- (B) $\exists x \in \mathbb{R}$ such that $x \geq 100$.
- (C) f is a polynomial function and f is not continuous at 0.
- (D) f is a polynomial function and f is not continuous anywhere.
- (E) $\exists x \in \mathbb{R}$ such that $\forall y \in \mathbb{R}, y \neq x^3$.
- (F) $\forall$ real-valued functions $f(x)$, $f(x)$ is continuous at any real number x .

Question 7

Conjecture: The sum of two even integers is divisible by 4.

- (A) For all integers x, y , if x and y are even then $x + y$ is divisible by 4.
- (B) x, y are even integers and $x + y$ is not divisible by 4.
- (C) Let $x = 2$ and $y = 0$. In this case $x + y = 2 + 0 = 2$, which is not divisible by 4. Therefore the conjecture is false.

Section 1.2

Question 5

- (A) x and y are real numbers such that $xy = 0$ and $x \neq 0$ and $y \neq 0$.
- (B) $\exists x \in \mathbb{Z}$, x^2 is not odd or $x^3 - 1$ is not divisible by 4.
- (C) $\exists n \in \mathbb{Z}$ such that $\forall k \in \mathbb{Z}$, $n \neq 2k \wedge n \neq 2k + 1$.
- (D) $\forall r \in \mathbb{R}$, $r \leq 1 \wedge r \geq 2$.
- (E) $\forall r \in \mathbb{R}$, r is less than or equal to 2 and greater than or equal to 1.
- (F) All functions are differentiable at 0 and continuous at 0.

Question 6

Conjecture: Every multiple of 6 is even and is not a multiple of 4.

- (A) $\forall n = 6a \mid n = 2k \wedge n \neq 4b$. $\{n, a, k, b \in \mathbb{N}\}$.
- (B) $\exists n = 6a \mid n \neq 2k \vee n = 4b$. $\{n, a, k, b \in \mathbb{N}\}$
- (C) Counterexample: $12 = 6 \cdot 2 = 4 \cdot 3$.

Section 1.3

Question 1

- (A) $P : \forall \text{ Hexagons } H, H \text{ has 6 sides.}$
 $\sim P : \exists H \mid H \text{ does not have 6 sides.}$
- (B) $P : \forall n, a, b \in \mathbb{N} \ n = 2a \vee n = 2b + 1$.
 $\sim P : \exists n, a, b \in \mathbb{N} \mid n \neq 2a \wedge n \neq 2b + 1$.
- (C) $P : \forall f$ that is differentiable at 0, f is continuous at 0.
 $\sim P : \exists f$ that is differentiable at 0 such that f is not continuous at 0.
- (D) $P : \forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $\sqrt{x} = y$.
 $\sim P : \exists x \in \mathbb{R}$ such that $\forall y \in \mathbb{R}, \sqrt{x} \neq y$

Question 7

Conjecture: The sum of the squares of three consecutive integers is even.

- (A) $\forall x, a \in \mathbb{N}, x^2 + (x + 1)^2 + (x + 2)^2 = 2a$. $\{a, x \in \mathbb{N}\}$.
- (B) $\exists x, a \in \mathbb{N} \mid x^2 + (x + 1)^2 + (x + 2)^2 \neq 2a$. $\{a, x \in \mathbb{N}\}$.
- (C) Counter-example: Let $x = 0 \therefore 0^2 + 1^2 + 2^2 = 5, 5 \neq 2a$.

Section 1.4

Question 4

- (A) **Contrapositive:** If the food didn't spoil, then the power did not go off.
Converse: If the food spoils, then the power went off.
- (B) **Contrapositive:** If the door is locked, then the light is not on.
Converse: If the door is not locked, then the light is on.
- (C) **Contrapositive:** If we didn't start the test, then it is not 9 A.M.
Converse: If we started the test, then it is 9 A.M.

Question 5

Statement: If $x^2 - x - 2 = 0 \rightarrow x = -1 \vee x = 2$.

- (A) **Contrapositive:** If $(x \neq -1 \wedge x \neq 2) \rightarrow x^2 - x - 2 \neq 0$.
Converse: If $(x = 1 \vee x = 2) \rightarrow x^2 - x - 2 = 0$.

Question 7

Statement: If f is differentiable at 0, then f is continuous at 0.

- (A) **Contrapositive:** If f is not continuous at 0, then f is not differentiable at 0.
Converse: If f is continuous at 0, then f is differentiable at 0.